

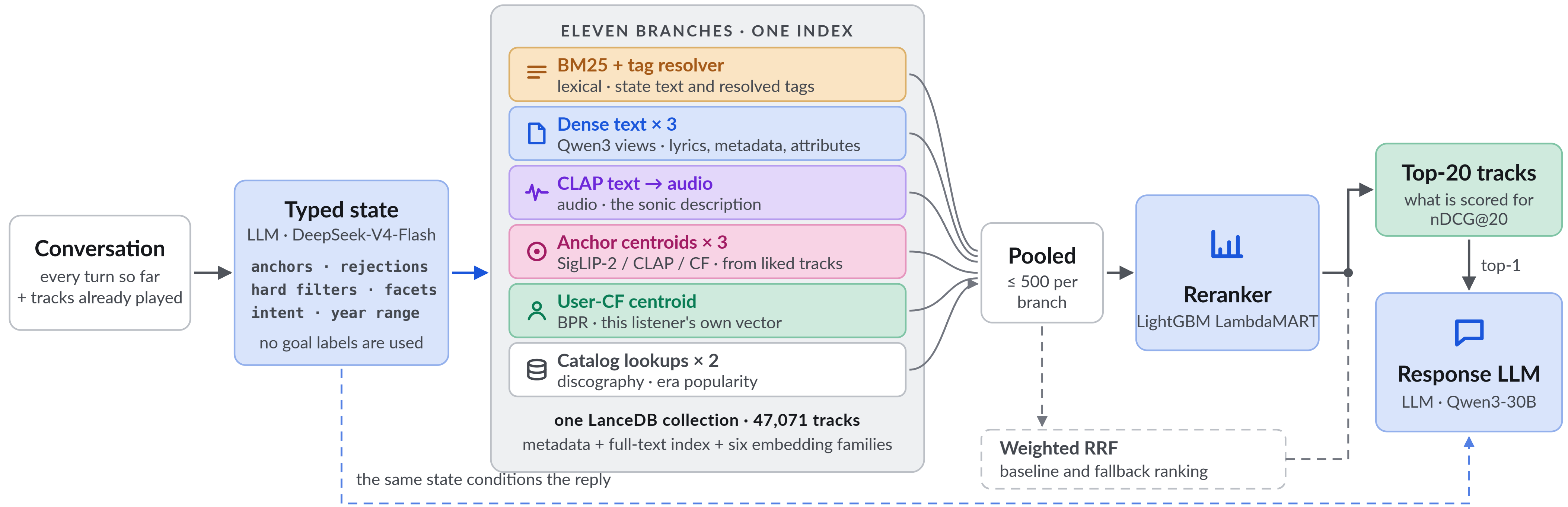
State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11 retrieval branches 146 ranker features 29/40 where that finished



Paper, code, models, data
github.com/npatta01/music-crs-2026



What the system reads off a conversation

Earlier turns: we played Neko Case's "Calling Cards" ("not quite the one"), then "Man" ("not it either").

User, now: "...much more subdued... a specific sense of place in the lyrics... a somewhat stark, almost a cappella delivery."

EXTRACTED STATE

anchor artist Neko Case (must use) • earlier turns
rejected "Calling Cards", "Man" • earlier turns
facets mood = subdued • theme = sense of place
sonic = stark, almost a cappella
request hidden_target

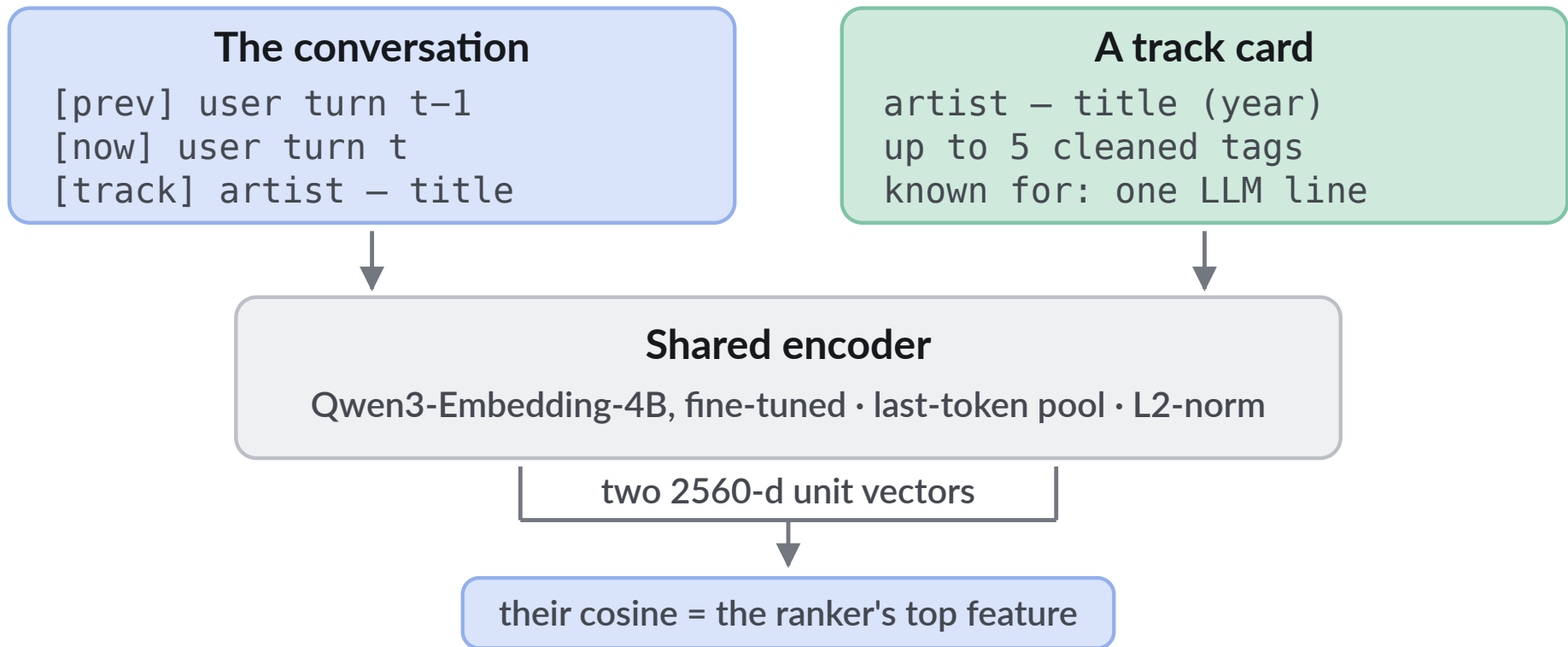
The full schema: request type • intent mode • anchors and pinned tracks • per-track feedback • rejected tracks and artists • hard filters • mood, genre and sonic facets • process constraints • routing flags • lyrical theme • release-year window.

What the ranker leans on

FEATURE FAMILY	SHARE OF GAIN
Bi-encoder cosine	59.0%
Per-branch rank and score	11.5%
Session and artist affinity	10.7%
Popularity • state • tags	9.2%
Other similarity cosines	8.9%
Constraints and rejections	0.7%

LightGBM LambdaMART • 146 features per (turn, track) pair.

THE BI-ENCODER BEHIND THAT TOP FEATURE



How it scored

OFFICIAL SPLIT	NDCG@20	JUDGE	COMPOSITE
Blind-A • development phase	0.4380	4.70	0.5799
Blind-B • final, 29th of 40	0.2537	3.30	0.3811

What is not ideal

We tuned on what we measured.

The development set was both our training and our evaluation data, and the response prompt was picked against the Blind-A leaderboard: 4.70 there, 3.30 on Blind-B.

Nothing from outside the catalog.

No outside source for what a track or artist is known for, and no tempo, key or soundtrack. Past one CF vector, no listening behaviour.

We extracted more than we used.

Most fields in the state never reached retrieval or ranking at all.

The reply was never checked.

Asked for a track we do not have, it described another by the same artist.

Asked for, served: where our top pick missed

THE LISTENER ASKED FOR

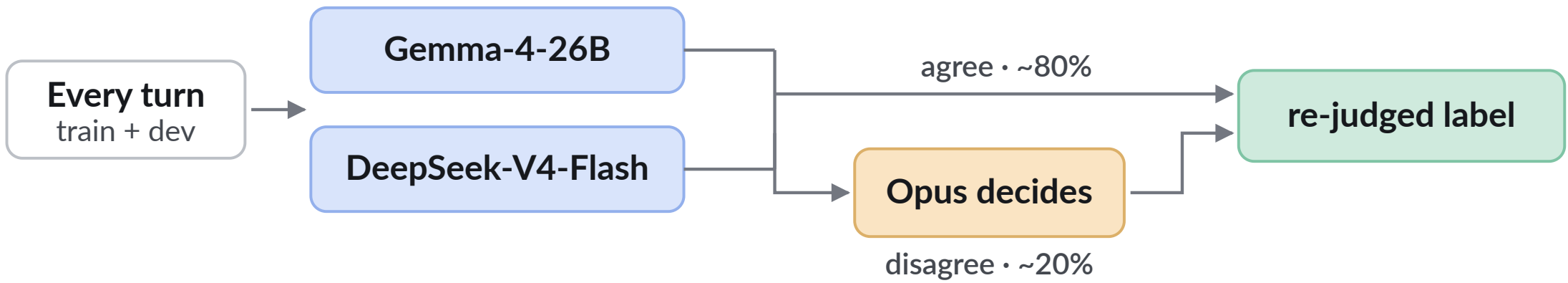
the Kamelot track from <i>Silverthorn</i>	→	Kamelot — Fallen Star, from <i>Haven</i> CONSTRAINT NEVER APPLIED
exactly 126.70 bpm, in G minor	→	no tempo or key field exists to match on NOTHING IN THE CATALOG
"Watercolors" by Pat Metheny	→	Metheny — Alfie, described as if it were RESPONSE LLM

The training labels often ignore the request

User: "...similar chill vibe, but from a different artist?" (just played: Bonobo)
Target: **Bonobo** — "Jets" • annotated MOVES toward goal
Next turn: "...hoping for something from a different artist this time..."
Target: **Bonobo** — "Pieces" • annotated MOVES toward goal

The label is whatever played next in an LLM-generated session, even when the listener had just asked for something else.

OUR FIX: TWO LLM JUDGES RE-SCORE EVERY TRAINING TURN



61%

of training turns where the judges would not settle on the target being a good fit

17%

keep the artist the listener had just rejected

Did it help? Retrained on these labels, it scored lower against the originals, so we kept the originals. That test could never favour the fix.